The running example for the metaplectic Jacquet–Langlands book: $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1,\, 4 - \zeta_3)_K$

David Joyner Claude (Anthropic)

November 8, 2026 — draft; resynchronized to the book November 9, 2026

Abstract

We fix one concrete case of the standing assumptions of [1] and [2], and evaluate in it the main objects of each chapter. These are the cocycle parameters, the commutator pairings, the good elements, the constants b_D and c_D, the wild toral decompositions, and the global normalization ledger. Every statement is either checked here, or quoted with its location and status label. The aim is to make the general constructions concrete, and to give each chapter a common place to test its formulas.

Theorem numbers. The four parts and the companion paper on the rank-two wild place have been combined into a single book [1], which is now the current source; references below are to its chapter, section and theorem numbers (Chapters 4–7 are Parts I–IV, and Chapter 8 is the companion paper). Numbers are current as of the book draft committed alongside this revision and can shift in later drafts, since the book is renumbered automatically when sections are added or removed.

1 The data

Base field	$K = \mathbb{Q}(\zeta_3)$, totally imaginary, $\mu_3 \subset K$
Covering degree, rank	$n = 3, \quad r = 2, \quad (n, r) = 1$
Parameters (Assumption 4.1.1)	$r' = 2, \quad c = 1, \quad c_v = r'c = 2$
Checks	$3 \mid r - 1 + 2c = 3, \quad 3 \mid r - 1 + 2rc_v = 9$
Division algebra	$D = (-1,\, 4 - \zeta_3)_K, \quad 4 - \zeta_3 = (1 - \zeta_3)(3 + \zeta_3)$
Ramification	$S = \{\mathfrak{p}_3, \mathfrak{p}_7\}, \quad \mathfrak{p}_3 = (1 - \zeta_3), \quad \mathfrak{p}_7 = (3 + \zeta_3)$
Cyclic subfield ([2])	$L = K(i) = \mathbb{Q}(\zeta_{12}) \subset D$

Fact 1 (**(P)** Ramification)**.** D is ramified exactly at $\mathfrak{p}_3$ and $\mathfrak{p}_7$.

Proof. At $\mathfrak{p}_3$ and $\mathfrak{p}_7$. The element -1 is a nonsquare in the residue fields $\mathbb{F}_3$ and $\mathbb{F}_7$, since $3 \equiv 7 \equiv 3 \pmod 4$. The element $4 - \zeta_3$ has valuation 1 at each of these places. So $(-1, 4 - \zeta_3) = -1$ at both.

At the other finite odd places. Both entries are units, so the symbol is 1. This includes the other prime above 7, namely $(3 + \zeta_3^2)$, since 7 splits in K.

At 2. The product formula forces the symbol to be 1.

Machine check. Sage gives:

```
K.<z> = CyclotomicField(3); a, b = K(-1), 4 - z
for p in [2,3,5,7,13]:
    for P, e in K.ideal(p).factor():
        print(p, P, K.hilbert_symbol(a, b, P))
```

This returns -1 exactly at $(z-1)$ and at $(-3z-2)$, and $+1$ at (2), (5), $(3z+1)$, $(z+4)$ and $(z-3)$. Here $(-3z-2) = (3+z)$, since $(-3z-2)/(3+z) = z^2$ is a unit. □

The hypotheses of Assumption 4.1.1 (Chapter 4, continuing those of [2, §1]) also hold.

- $L = K(i)$ is cyclic of degree $r = 2$ and lies in D.
- At both places of S, L_v is a field, because -1 is not a square in $\mathbb{F}_3$ or $\mathbb{F}_7$.
- $|S| = 2$ is even, as it must be for $r = 2$.

A fuller check of this example against the hypotheses of [2], [3] and [4] is in the book's Chapter 1 (Example 1.3.1 and Remark 1.3.2).

2 The local fields

Place	K_v	Type for $n = 3$	$\lvert 3\rvert_v$	$\lvert K_v^\times / K_v^{\times 3}\rvert$
$\mathfrak{p}_3 \in S$	$\mathbb{Q}_3(\zeta_3)$ ($e = 2$, $q = 3$)	wild	$1/9$	81
$\mathfrak{p}_7 \in S$	$\mathbb{Q}_7$ (contains μ_3)	tame	1	9
(2)	$\mathbb{Q}_2(\zeta_3)$ ($f = 2$, $q = 4$)	tame, split for D	1	9
$(3+\zeta_3^2)$, $(13) = \mathfrak{p}\mathfrak{p}'$, …	$\mathbb{Q}_\ell$, $\ell \equiv 1$ (3)	tame, split	1	9
complex	$\mathbb{C}$	cover splits	9	1

The last column is $n^2|n|_v^{-1}$ at the finite places (Chapter 7, Proposition 7.4.2). Locally at $\mathfrak{p}_3$ we have $3+\zeta_3 \equiv 1 \pmod{\mathfrak{p}_3}$, which is a square. So $D_{\mathfrak{p}_3} \cong (-1, \varpi)$ with $\varpi = \zeta_3 - 1$; this is exactly the model $E_0 \oplus E_0 j$, $j^2 = \varpi$, of the companion chapter, Chapter 8, Example 8.7.1. At $\mathfrak{p}_7$, $D_{\mathfrak{p}_7} \cong (-1, 7)_{\mathbb{Q}_7}$.

3 Chapter 4: cocycle, commutators, good elements

- *Diagonal torus* (Chapter 7, Proposition 7.2.1). With $c_v = 2$,

$$[\tilde{t}, \tilde{u}] = \prod_{i,j} (t_i, u_j)_3^{A_{ij}}, \qquad A = -I + 5J \equiv \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \pmod 3,$$

  so $[\tilde{t}, \tilde{u}] = (t_1/t_2, u_1/u_2)_3$. A regular $\mathrm{diag}(t_1, t_2)$ is good if and only if t_1/t_2 is a cube.

- *Quadratic tori* (Chapter 4, Proposition 4.3.1 and Example 4.3.7). Here $[x, y] = (Nx, Ny)_F^2 (x, y)_E^{-1}$. The good elements are $F^\times E^{\times 3}$ (Chapter 4, Proposition 4.3.5).

- *Index of the good elements in a quadratic torus*, $|3|_v^{-1}$ (Chapter 4, Proposition 4.3.5):
  - at $\mathfrak{p}_3$ it is 9: the covered torus is a Heisenberg group, and its genuine irreducibles have dimension 3;
  - at $\mathfrak{p}_7$ it is 1: every element is good, and the covered torus is abelian.

- *Sign.* The matching sign $(-1)^{r-1}$ of Chapter 4, Theorem 4.4.5 (Theorem B), is -1.

- *A regular-but-not-good element* (Chapter 7, Remark 7.3.3). The element $\mathrm{diag}(1, \zeta_3)$ is regular. It is not good at $\mathfrak{p}_3$, because $\zeta_9 \notin \mathbb{Q}_3(\zeta_3)$. Its cube is the identity.

4 Chapter 6: the transform L_D

	$b_D = \lvert 3\rvert^{-1/2}$	$c_D = \lvert 3\rvert^{-3/2}$	$R_E = \Delta(t)/\Delta(t^3)$
$\mathfrak{p}_7$ (tame)	1	1	$\equiv 1$ (Chapter 6, Lemma 6.7.4)
$\mathfrak{p}_3$ (wild)	3	27	$\in \{1, 3, 9\}$ (below)

- *At $\mathfrak{p}_7$.* L_D is a dimension-preserving bijection of irreducible characters (Chapter 6, Theorem 6.7.6) **(P)**.
- *At $\mathfrak{p}_3$, divisibility.* The cover does not split, and every genuine irreducible has dimension divisible by 3 (Chapter 6, Proposition 6.8.1) **(P)**.
- *At $\mathfrak{p}_3$, the prediction.* If L_D maps irreducibles to irreducibles, then $\dim \tau = 27 \dim \rho$. In particular $L_D\mathbf{1}$ would be an irreducible of dimension 27.

5 Chapters 7 and 8: the wild place $\mathfrak{p}_3$

The three quadratic tori

$F = \mathbb{Q}_3(\zeta_3)$ has three quadratic extensions. All of them embed in $D_{\mathfrak{p}_3}$. On $E^1 \cong E^\times/F^\times$, Chapter 7's toral-positivity analysis gives the following **(P)**.

E/F	values of R_E	R_E as a permutation character	coefficients
unramified $F(i)$	9, 1	$1 + 2\,\mathrm{Reg}(C_4)$	$(3, 2, 2, 2)$
ramified $F(\sqrt{u\varpi})$, either type	9, 3, 1	$1 + \mathrm{Reg}(C_2) + \mathrm{Reg}(C_6)$	$(3, 2, 1, 1, 1, 1)$

Both expansions have value $9 = |3|_F^{-1}$ at the identity. So the restriction of $L_D\mathbf{1}$ to every covered quadratic torus is an actual genuine character of dimension $3 \cdot 9 = 27$, consistent with the prediction above. In the cyclic algebra model $j^2 = \varpi$:

t	j	$1 + j$	$1 + \varpi j$
$R(t)$	1	3	9
$(L_D\mathbf{1})(t^*)$	3	9	27

What the book now proves

This question is no longer open for the running example. Chapter 8 (the companion chapter) proves, as its main theorem (Target A) **(P)**: $L_D\mathbf{1} = \Theta_\tau$ for a unique genuine irreducible representation τ of $\widetilde{D^\times_{\mathfrak{p}_3}}$, of dimension 27, exhibited explicitly as induced from a canonical genuine character of the third congruence subgroup. The proof runs through hypothesis (H1) (Chapter 8, Proposition 8.11.4), the theorem that (H1) implies Target A (Chapter 8, Theorem 8.10.6), and the uniqueness/irreducibility of the restriction to the pro-3 part (Chapter 8, Theorem 8.9.5). The same chapter extends this to every representation of $D^\times$ trivial on the third congruence subgroup and its norm twists. What remains open – rows of higher level in this example, and the wild place for other primes and base fields – is discussed in the book's Chapter 8, §8.15 (Future directions). This is no longer the smallest open instance of the wild integrality problem; it is the first solved one.

6 Global: good classes and the normalization ledger

- *Good classes* (Chapter 7, Theorem 7.5.2) **(P)**. A regular $\gamma \in D^\times(K)$ with $E = K(\gamma)$ has a rational lift that is good everywhere if and only if $\gamma \in K^\times E^{\times 3}$. The map $[s] \mapsto [s^3]$ is a bijection onto these classes.
- *The ledger* (Chapter 7, Example 7.8.8). Take $\lambda_v = 1$ at the finite split places. Then

$$\Lambda = \lambda_\infty \cdot b_{D,\mathfrak{p}_3} \cdot b_{D,\mathfrak{p}_7} = 3\lambda_\infty,$$

  and $\Lambda = 1$ forces $\lambda_\infty = 1/3 = |3|_\infty^{-1/2}$.
- *The identity-term heuristic.* It gives the complex factor $\lambda_\infty |3|_\infty^{-1} = 1/27$, which balances $c_{D,\mathfrak{p}_3} = 27$ by the product formula.
- *Unramified places.* [3]'s unit matching needs $p \nmid n$ and $(n, N_2) = (3, 1) = 1$. Both hold at every finite place outside S, since $\mathfrak{p}_3 \in S$ is the only place above 3. (As the book's Chapter 1, Remark 1.3.2 notes, $(n, N_2) = 1$ holds automatically for every odd n in the quaternion case $r = 2$, since there are no composite integers at most 2.)

7 Relation to [2]

In [2] the lifting runs from genuine automorphic representations of an n-fold cover of $L_{\mathbb{A}}^\times$, here with $L = \mathbb{Q}(\zeta_{12})$, to ε-invariant genuine representations of the cover of $D_{\mathbb{A}}^\times$. Chapter 4, §4.1, of the book says this. A section of [2] evaluated in this example would complete the picture.

Suggested use

Each chapter of the book has a short subsection, "A running example" (or, in Chapter 1, the canonical statement, Example 1.3.1), pointing to the relevant section of this note; [1] cites this note in the normal way wherever the example is worked out further. Each new formula in a draft can be tested here first. Chapter 8, §8.15 is the natural place to look for the next target in the wild integrality problem, now that this example's own wild place is settled.

References

[1] D. Joyner, Claude and ChatGPT, *On a metaplectic Jacquet–Langlands transfer for division algebras of prime degree*, book draft, `book/mjl_book_draft.tex`, this repository, 2026.

[2] D. Joyner and Claude, On the metaplectic analog of Kazhdan's "endoscopic" lifting (original 1988, corrected 2026), `metaplectic_kazhdan_revised12.tex`.

[3] Y. Z. Flicker and D. A. Kazhdan, Metaplectic correspondence, *Publ. Math. IHÉS* 64 (1986), 53–110.

[4] Y. Z. Flicker and D. A. Kazhdan, A simple trace formula, *J. Analyse Math.* 50 (1988), 189–200.